Element-Wise Femoral Stress Prediction: A Hybrid Graph Neural Network Surrogate for Finite Element Analysis

Sikha O K[1], **Abdollah Moslemi**[1], **Elham Alizadeh**[1], **Ludovic Humbert**[2], **Miguel A. González Ballester**[1,3] and **Jérôme Noailly**[1]

[1] *BCN MedTech, Department of Engineering, Universitat Pompeu Fabra, Barcelona, Spain, sikha.okkath@upf.edu*
[2] *3D-Shaper Medical, Barcelona, Spain*
[3] *ICREA, Barcelona, Spain*

Introduction

Hip fragile fractures are a major clinical burden, with high mortality, disability and rising incidence. Finite element analysis (FEA) provides accurate subject-specific assessment of femoral strength [1,2], but high computational costs limit routine clinical use. Machine learning (ML) surrogates can approximate FEA outputs at a fraction of the cost. Yet, conventional ML treats each mesh element independently, ignoring the spatial structure of the finite element (FE) mesh. Graph neural networks (GNNs) address this by operating directly on the mesh and propagating information among connected elements. We propose a hybrid GCN–SAGE–Residual (Graph Convolutional Network with Graph Sample and Aggregate) GNN that predicts element-wise stress in the proximal femur from FE mesh representations, and benchmark it against classical ML and single-operator GNN.

Methods

A cohort of 128 women (64 incident hip fractures, 64 age-matched controls) was analysed. Patient-specific hexahedral FE models of the proximal femur (126,800 elements) were generated from 3D-DXA via statistical-shape-model morphing, and a sideways-fall load onto the greater trochanter was simulated in Abaqus [1], yielding the six Cauchy stress components per element as the ground truth to be targeted. Each mesh was converted into a graph in which each element is a node and face-sharing information between elements are edges. Nodes carried nine descriptors (centroid coordinates; apparent density; elastic modulus; tissue type; anatomical region; graded boundary-condition and loading labels). The proposed architecture fuses graph convolution (GCN), GraphSAGE aggregation, and residual connections in every layer (six layers, 256 channels). It is trained with a stress-weighted MSE loss emphasising fracture-prone regions. Models were evaluated on unseen test cases (R^2, MAE, RMSE) and by high-stress hotspot detection (top-10% threshold).

Results

GNN models substantially outperformed classical ML (Table 1). Linear models were poor ($R^2 \approx 0.10$, MAE > 6 MPa) and the best classical model, Gradient Boosting, reached $R^2 = 0.76$ but with high error (MAE = 3.25 MPa). The proposed hybrid model achieved the best overall performance ($R^2 = 0.7987$, MAE = 0.1891 MPa, RMSE = 0.4472), improving on the strongest single-operator GNN baseline (GCN-256, $R^2 = 0.7558$) by 5.7% in R^2. R^2 for each stress component ranged from 0.808 to 0.889, indicating balanced accuracy for both normal and shear components (Fig. 1A).

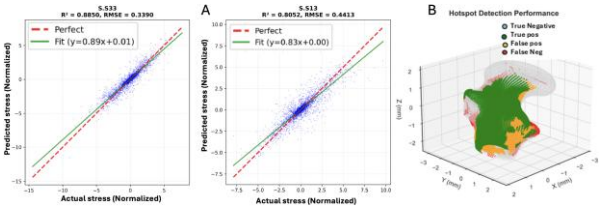


Fig. 1: Performance of the GCN-SAGE-Residual (hybrid) model: A) Scatter plots for the 33 and 23 stress component prediction (R^2) and B) Mechanical prediction accuracy in regions of prone to fracture.

Reformulated as hotspot detection, the model localised high-stress regions with Precision 0.801, Recall 0.785 and F1-score 0.793, showing strong spatial overlap with FEA-derived failure-prone zones (Fig. 1B).

Conclusions

A hybrid GCN–SAGE–Residual GNN predicts element-wise proximal-femur stress directly from FE meshes, outperforming both classical ML and single-operator GNN baselines and preserving the full directional stress tensor. Combining complementary aggregation operators with residual connections, rather than scaling any single operator, is the key to accuracy. Reliable hotspot localisation makes the surrogate a fast, clinically interpretable alternative to FEA for fracture-risk assessment. Future work will address multiple loading scenarios, other anatomical sites, and integration with CT-based clinical pipelines.

Acknowledgements

European Research Council (ERC-2021-CoG-O- Health-101044828); Spanish Ministry CPP2023-010970; AGAUR (2023 BP 00224)

References

1. Ruiz Wills et al., Bone, 121:89-99, 2019.
2. Alizadeh et al., Front Bioeng Biotechnol, 13, 2025.

Table 1: R^2 and MAE for all evaluated models.

Metric	Ridge	Lasso	Elastic Net	Random Forest	Gradient Boosting	CatBoost	MLP	GCN-128	SAGE-128	GAT-128	GCNResidual-128	GCN-256	SAGE-256	**GCN-SAGE-Residual**
R^2	0.105	0.102	0.098	0.762	0.763	0.752	0.748	0.731	0.731	0.683	0.733	0.756	0.754	**0.799**
MAE (MPa)	6.963	6.888	6.899	3.177	3.246	3.394	3.370	0.216	0.218	0.238	0.215	0.204	0.206	**0.189**